

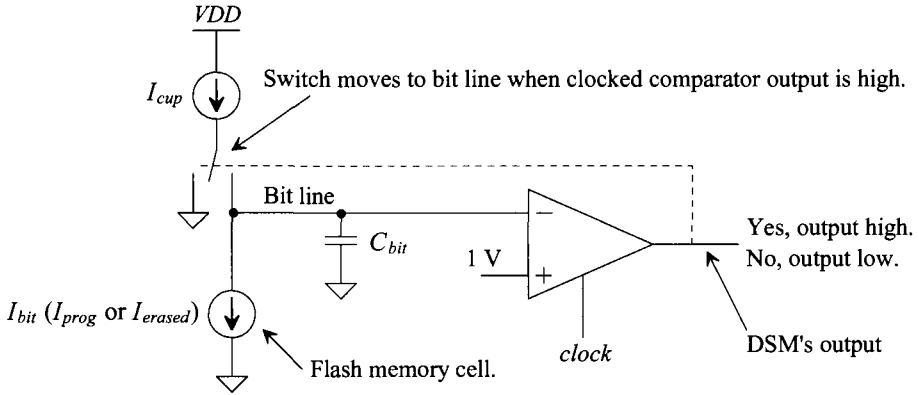


Figure 17.6 Sensing a Flash memory cell using DSM.

the current source, I_{cup} , is connected to C_{bit} to provide the water to the bucket. Note that *we cannot connect the bit line to a voltage source when the output of the comparator goes high*. A voltage source, in our water analogy, is a pressure with an infinite supply of water. The connection of the voltage source would fill the bucket up (the bit line capacitance) to the pressure of the source (the voltage of the source). Again, by looking at the number of times the output of the DSM sensing circuit goes high, we can determine, very precisely, the value of I_{bit} .

If we are clocking the comparator at a rate of f at a period of $T (= 1/f)$, then the rate that charge is removed from the bit line capacitance (the sigma bucket) is

$$\frac{I_{bit}}{C_{bit}} = \frac{\Delta V_{bit}}{T} \quad (17.3)$$

where ΔV_{bit} is the change in the bit line voltage. The amount of charge removed from the bit line in one clock cycle T is

$$Q_{bit} = I_{bit}T = C_{bit} \cdot \Delta V_{bit} \quad (17.4)$$

These quantities are constant. However, the rate at which we add charge to the bit line isn't a constant (unless the output of the comparator is always the same). If N is the total number of clock cycles (the total number of times we clock the comparator) and M is the number of times the output of the comparator goes high (the number of "Yes, add charge to the capacitor" signals), then the rate we add charge to the bit line from I_{cup} is

$$Q_{cup} = I_{cup} \cdot \frac{M}{N} \cdot T \quad (17.5)$$

In order for the voltage on the bit line (the water level in the bucket) to remain, on average, constant, we require that the amount of charge leaving the bucket (Q_{bit}) equal the amount of charge entering the bucket (Q_{cup}) or

$$Q_{bit} = I_{bit}T = Q_{cup} = I_{cup} \cdot \frac{M}{N} \cdot T \quad (17.6)$$

or

$$\frac{I_{bit}}{I_{cup}} = \frac{M}{N} \quad (17.7)$$

This result is important. It relates the counter output code, M , and the total number of times the DSM sensing circuit is clocked, N , to the ratio of the Flash memory cells current, I_{bit} and the fed-back signal I_{cup} . Note again that the charge leaving the bit line capacitance can't be greater than the charge entering the bit line capacitance, that is, we require $I_{cup} \geq I_{bit}$. As mentioned earlier, using a large cup (large I_{cup}) increases the sensing time for a required resolution (we're averaging a larger variable).

Notice that Eq. (17.7) doesn't include the clock frequency or period. We might think that these quantities aren't important. While, if the DSM is designed correctly, it doesn't directly affect the output of the sensing operation, we can have the situation where the bucket (bit line capacitance) empties or overflows (goes to ground or V_{DD}). To avoid this situation, we may require that the maximum deviation on the bit line, $\Delta V_{bit,max}$ be less than some value over a time T . We can write, assuming $I_{cup} \geq I_{bit}$,

$$\Delta V_{bit,max} = \frac{I_{cup}T}{C_{bit}} = \frac{I_{cup}}{C_{bit} \cdot f_{clk}} = I_{cup} \cdot R_{sc} \quad (17.8)$$

For example, if the bit line capacitance is 500 fF, the clock frequency is 100 MHz ($T = 10$ ns), and I_{cup} is 10 μ A, then $\Delta V_{bit,max} = 0.2$ V. If we clock the DSM slower, we must increase our bucket size (add capacitance in parallel with C_{bit} on the input of our DSM).

Example 17.1

Suppose that the DSM sensing circuit in Fig. 17.6 is used to determine the current flowing in a programmed Flash memory cell. If the single transistor Flash memory cell is programmed to conduct 1, 3, 5, or 7 μ A of current, estimate the counter output codes if the DSM is clocked 15 times (a 4-bit counter is used) and I_{cup} is 10 μ A. Estimate the maximum bit line voltage change if the DSM is clocked at 100 MHz and the bit line capacitance is 500 fF.

For the 1 μ A program current, using Eq. (17.7), we get

$$\frac{1}{10} = \frac{M}{15} \rightarrow M = 1.5 \text{ so the counter output would be 2 (0010)}$$

For the 3 μ A program current through the Flash memory cell,

$$\frac{3}{10} = \frac{M}{15} \rightarrow M = 4.5 \text{ so the counter output would be 5 (0101)}$$

For the 5 μ A program current,

$$\frac{5}{10} = \frac{M}{15} \rightarrow M = 7.5 \text{ so the counter output would be 8 (1000)}$$

Finally, for the 7 μ A program current,

$$\frac{7}{10} = \frac{M}{15} \rightarrow M = 10.5 \text{ so the counter output would be 11 (1011)}$$

The maximum deviation of the voltage on the bit line would be when the program current is 1 μ A (leaving the bit line) and I_{cup} is connected (10 μ A charging the bit line). This would give a net current of 9 μ A into the bit line so

$$\Delta V_{bit,max} = \frac{9\mu A \cdot 10\text{ ns}}{500\text{ fF}} = 180\text{ mV}$$

Let's show the output decisions for the case when I_{prog} is $1\text{ }\mu\text{A}$, Table 17.2. Note that if $1\text{ }\mu\text{A}$ is removed from the 500 fF bit line capacitor for 10 ns , the bit line voltage drops 20 mV . If (net) $9\text{ }\mu\text{A}$ of current is put into the bit line for 10 ns , the bit line voltage increases by 180 mV . Note that as calculated, the maximum change in the bit line voltage is 180 mV . Also note that if the comparator had output a Yes at 100 ns instead of at 110 ns , it wouldn't have affected the final output. ■

Table 17.2 See Ex. 17.1.

Time, nanoseconds	Bit line voltage	Add current? ($V_{bit} < 1\text{ V}$?)	Bit line current	Counter output
0	1		$-1\text{ }\mu\text{A}$	0 (0000)
10	0.98	Yes (1)	$9\text{ }\mu\text{A}$	1 (0001)
20	1.16	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
30	1.14	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
40	1.12	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
50	1.1	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
60	1.08	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
70	1.06	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
80	1.04	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
90	1.02	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
100	1	No (0)	$-1\text{ }\mu\text{A}$	1 (0001)
110	0.98	Yes (1)	$9\text{ }\mu\text{A}$	2 (0010)
120	1.16	No (0)	$-1\text{ }\mu\text{A}$	2 (0010)
130	1.14	No (0)	$-1\text{ }\mu\text{A}$	2 (0010)
140	1.12	No (0)	$-1\text{ }\mu\text{A}$	2 (0010)
150	1.1	No (0)	$-1\text{ }\mu\text{A}$	2 (0010)

Notice, in the previous example, that we can clock the DSM indefinitely. The only limitation is the size of the counter (after a time the counter will roll over). This is important because, to get better resolution, all we have to do is sense for a longer period of time. We can define the *dynamic range* of the sense operation, in dB, as the maximum counter output code (N) to the smallest output code (1)

$$\text{dynamic range (DR)} = 20 \cdot \log \frac{N}{1} = 20 \cdot \log N \quad (17.9)$$

noting that no signal would give a counter output code of zero. Clocking the DSM 1,000 times results in, ideally, a DR of 60 dB . Clocking the DSM 15 times results in a DR of 23.5 dB .

The maximum input signal, $I_{bit,max}$, equals the fed-back signal I_{cup} ($I_{cup} \geq I_{bit}$). The minimum resolvable signal is then

$$\text{minimum resolvable signal} = \frac{\text{fed-back signal}}{N} \quad (17.10)$$

Again, this illustrates that the larger the fed-back signal, the more averages, N , are needed for a given resolution.

Example 17.2

Determine the minimum resolvable programmed current, I_{bit} , in Ex. 17.1. Verify the answer with a table similar to Table 17.2.

Using Eq. (17.10), the minimum resolvable current is $10 \mu\text{A}/15$ or $0.666 \mu\text{A}$. If this current is removed from the bit line capacitance for 10 ns, the voltage drop is 13.33 mV. If the bit line capacitance is charged with $9.333 \mu\text{A}$ for 10 ns, its voltage increases to 186.66 mV. Table 17.3 tabulates the operation of the DSM sensing circuit when I_{prog} is 666.6 nA. Note that in all situations where I_{prog} is not zero the first decision is a Yes. However, after neglecting this decision, N clock cycles later the counter increments to 2 when the cell is sinking the minimum resolvable signal (here 666.6 nA). ■

Table 17.3 See Ex. 17.2.

Time, nanoseconds	Bit line voltage (millivolts)	Add current? ($V_{bit} < 1 \text{ V}$?)	Bit line current	Counter output
0	1,000		$-0.666 \mu\text{A}$	0 (0000)
10	986.67	Yes (1)	$9.333 \mu\text{A}$	1 (0001)
20	1,173.33	No (0)	$-0.666 \mu\text{A}$	1 (0001)
30	1,160	No (0)	$-0.666 \mu\text{A}$	1 (0001)
40	1,146.67	No (0)	$-0.666 \mu\text{A}$	1 (0001)
50	1,133.34	No (0)	$-0.666 \mu\text{A}$	1 (0001)
60	1,120	No (0)	$-0.666 \mu\text{A}$	1 (0001)
70	1,106.67	No (0)	$-0.666 \mu\text{A}$	1 (0001)
80	1,093.33	No (0)	$-0.666 \mu\text{A}$	1 (0001)
90	1,080	No (0)	$-0.666 \mu\text{A}$	1 (0001)
100	1,066.67	No (0)	$-0.666 \mu\text{A}$	1 (0001)
110	1,053.33	No (0)	$-0.666 \mu\text{A}$	1 (0001)
120	1,040	No (0)	$-0.666 \mu\text{A}$	1 (0001)
130	1,026.67	No (0)	$-0.666 \mu\text{A}$	1 (0001)
140	1,013.33	No (0)	$-0.666 \mu\text{A}$	1 (0001)
150	1,000	No (0)	$-0.666 \mu\text{A}$	1 (0001)
160	986.67	Yes (1)	$9.333 \mu\text{A}$	2 (0010)

The Feedback Signal

The precision of a sense operation in a DSM is limited by how precisely current can be guided into or out of the bit line capacitance. The bit line capacitance can be thought of as integrating (an integrator, *sigma*) the difference (*delta*) between the fed-back signal (I_{cup} in Fig. 17.6) and the signal we are measuring (I_{bit}). If the amount of charge (current) steered into the bit line isn't consistent from one clock period to the next, errors will occur. To illustrate the possibility of errors, examine the circuit seen in Fig. 17.7. When the switch is connected to ground $C_{parasitic}$ (the parasitic capacitance on the output of the current source), is discharged. Now when the switch is connected to the bit line I_{cup} must charge both $C_{parasitic}$ and C_{bit} . Because the bit line voltage will be moving around and the switch may stay connected to the bitline for two or more consecutive cycles, errors in the sense will occur that limit the resolution. For example, the charge supplied to $C_{parasitic}$ may be $1.05 \cdot C_{parasitic}$ (a bit line voltage of 1.05V) in one clock cycle, while in a different clock cycle it may be $1.02 \cdot C_{parasitic}$. In our water analogy in Fig. 17.4 the amount of water in the cup effectively changes due to this parasitic. If the size of the bit line capacitance is large (or if capacitance is added to the input of the DSM to purposely increase this capacitance) or only a coarse sense is used (say 128 or less clock cycles), then $C_{parasitic}$ will probably not affect the sense in any significant way. Note that if the switch were connected to 1V in Fig. 17.7 instead of ground, the errors from $C_{parasitic}$ would be reduced.

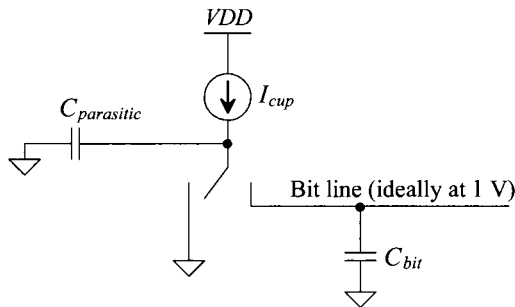


Figure 17.7 How the parasitic capacitance on the output of the current source causes errors.

In an attempt to minimize the power (power is burned unnecessarily when I_{cup} is connected to ground in Figs. 17.6 or 17.7) and the effects of parasitics consider the switched-capacitor (SC) circuit seen in Fig. 17.8. In the following we assume the bit line reference voltage (here 1 V) is greater than the threshold voltage of the PMOS transistors so that the PMOS switches used in the SC can turn fully on. The two clock signals, ϕ_1 and ϕ_2 form nonoverlapping clock signals (they are never low at the same time). This is important because we never want to connect the bit line directly to VDD (which occurs if the clock signals are all low at the same time). When ϕ_1 is low, C_{cup} charges to VDD . Next, ϕ_1 goes high. Between ϕ_1 going high and ϕ_2 going low, the comparator is clocked (on the rising edge of *clock*). If charge needs to be added to the bit line, the comparator output goes high and the gate of M3 is driven low. Note that the inverter can be removed

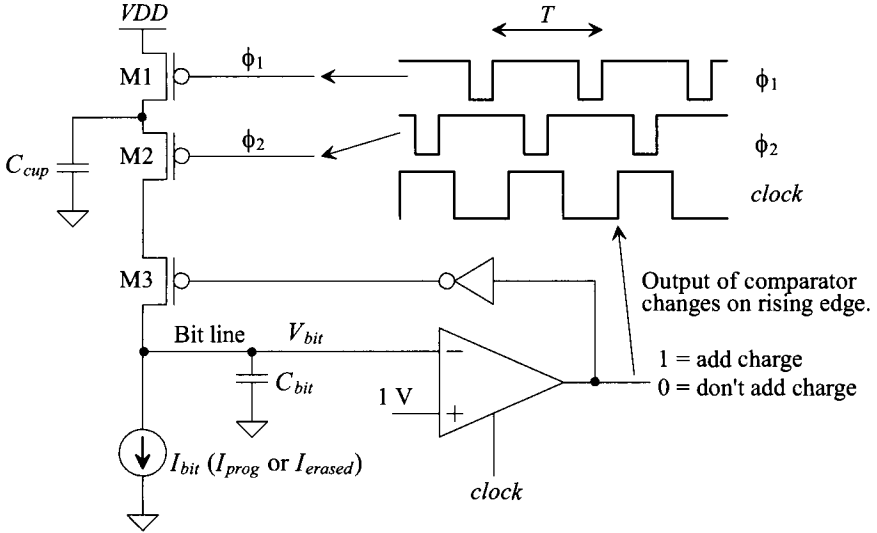


Figure 17.8 Using a switched-capacitor to add charge to the bit line.

if we swap the inputs of the clocked comparator so that a 1 output indicates “don’t add charge” and a 0 output indicates “add charge.” When ϕ_2 goes low, the charge on C_{cup} is dumped into the bit line. The charge leaving the bit line is still given by Eq. (17.4). The charge dumped into the bit line from C_{cup} is

$$Q_{cup} = C_{cup} \cdot (VDD - V_{bit}) \quad (17.11)$$

When we compare this amount of charge to Eq. (17.5), we see a benefit when Q_{cup} is not a function of the clock period. However, the big drawback is that the charge we add to the bit line is a function of the bit line voltage. This limits the accuracy of the sense. The water analogy to this problem is that the cup is being filled up to a level dependent on the sigma bucket water level. The amount of water dumped into the bucket from the cup should be independent of the water level in the bucket.

To make the charge we add to the bit line independent of the bit line voltage, consider the addition of another PMOS device in Fig. 17.9. When both the signal from the comparator and ϕ_2 are low, M2 and M3 are on. The charge from C_{cup} is dumped into the source of M4. The result is an initial increase in M4’s source potential (M4 turns on). However, after the charge is dumped into the bit line, M4 shuts off. Its V_{SG} goes to V_{THP} (M4’s source potential goes to $V_{REF} + V_{THP}$). This keeps the potential across C_{cup} constant. Equation (17.11) can be rewritten as

$$Q_{cup} = C_{cup} \cdot (VDD - V_{REF} - V_{THP}) \quad (17.12)$$

knowing, to keep M4 from being fully on, that is, $V_{SD,sat}$ not zero

$$V_{REF} + V_{THP} > V_{bit,max} \quad (17.13)$$

where $V_{bit,max}$ is the maximum voltage on the bit line (and, again, $V_{bit,min} > V_{THP}$).

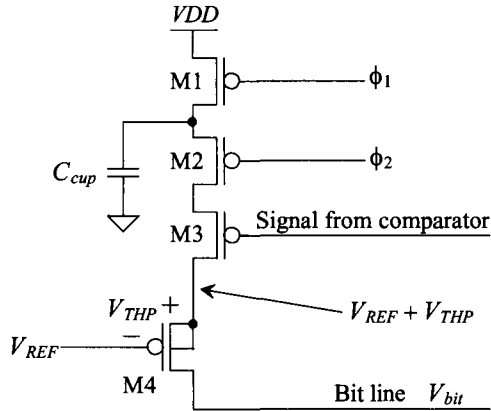


Figure 17.9 Adding a MOSFET to set the swing across the capacitor.

Example 17.3

Using SPICE demonstrate the validity of Eqs. (17.11) and (17.12).

To demonstrate the validity of Eq. (17.11), examine the circuit in Fig. 17.10. We know that a PMOS device passes a voltage from V_{DD} to V_{THP} , so we make sure that the bit line is always at a potential greater than V_{THP} ($= 280$ mV for the short channel process used in this book). If it's not, the PMOS devices don't behave like switches. In the simulation we'll sweep the bit line voltage from 0.3 to 1V and look at the current through the bit line voltage source, V_{bit} . M3 is always on so we can look at the SC's operation alone (without the effects of the comparator). If Eq. (17.11) is valid, we should see a linear decrease in the current pulses (the charge) as V_{bit} is increased. Figure 17.11a shows the nonoverlapping clocks used in the simulation. Note how neither clock (ϕ_1 or ϕ_2) is low at the same time (again, this is important). Figure 17.11b shows the current through V_{bit} . As expected it decreases as V_{bit} increases. As Eq. (17.11) indicates, the current goes to zero when

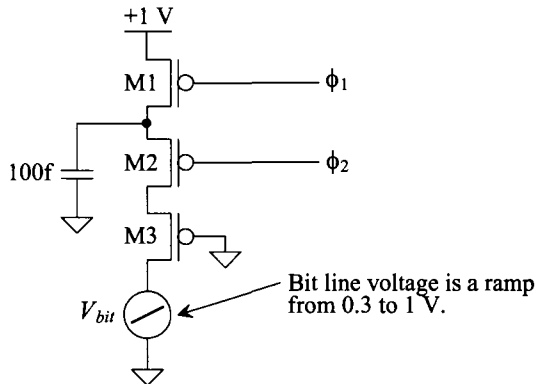


Figure 17.10 Verifying Eq. (17.11), see Ex. 17.3.

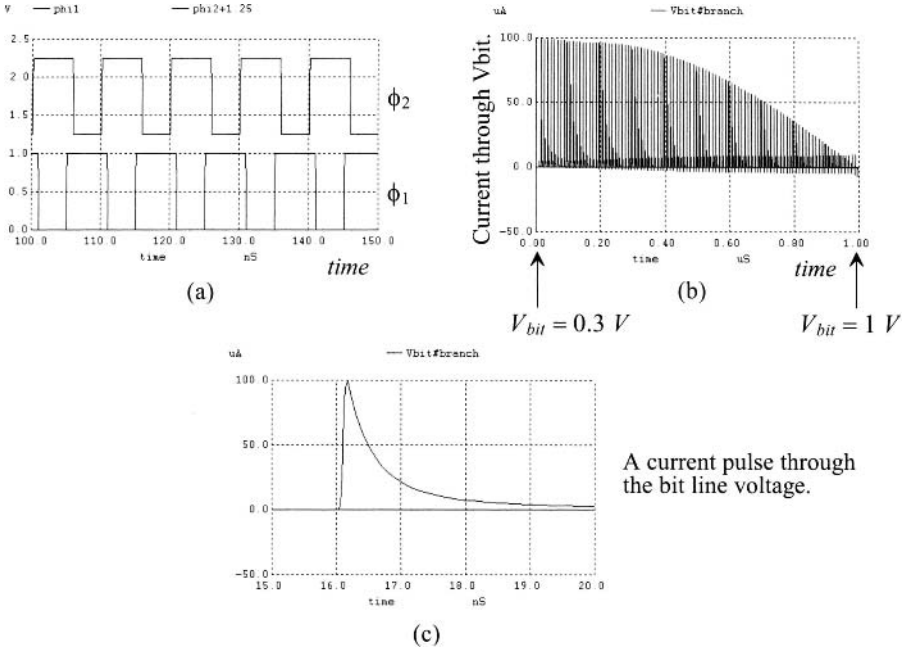


Figure 17.11 The operation of the circuit in Fig. 17.10.

V_{bit} is 1 V (= V_{DD}). Let's use Eq. (17.11) to calculate the charge supplied by C_{cup} when V_{bit} is 300 mV

$$Q_{cup} = 100 \text{ fF} \cdot (1 - 0.3) = 70 \text{ fC} \quad (17.14)$$

Figure 17.11c shows the pulse of current that is dumped into V_{bit} at the beginning of the simulation. To estimate the amount of charge in this pulse, we take the amplitude and multiply it by an estimate of the pulse's width. That is, $100 \mu\text{A} \cdot 0.7 \text{ ns} = 70 \text{ fC}$ or the same result as seen in Eq. (17.14).

To validate Eq. (17.12), we can add M4 from Fig. 17.9 to the circuit seen in Fig. 17.10. The only parameter we need to calculate before simulating is the value of V_{REF} . Using Eq. (17.13) and knowing $V_{THP} = 280 \text{ mV}$, we see that if we set V_{REF} to a large voltage, Q_{cup} gets small. If, for example, $V_{REF} = V_{DD} - V_{THP}$ (= 750 mV in this example), Q_{cup} goes to zero. We can write

$$V_{REF} < V_{DD} - V_{THP} \quad (17.15)$$

Let's use a V_{REF} of $V_{DD}/2$ or 500 mV. Figure 17.12 shows the simulation results. Note, in Fig. 17.9, that we place M4 in its own well. Since we care about how the threshold voltage varies, we've eliminated the body effect in this device. The simulation results are seen in Fig. 17.12. In this simulation we've swept the bit line voltage from 0.3 to 0.6 V. Comparing the plot to Fig. 17.11b, we see much better linearity (the charge added to the bit line doesn't change with bit line voltage). The cost for this improvement is more limited voltage swing on the bit line. ■

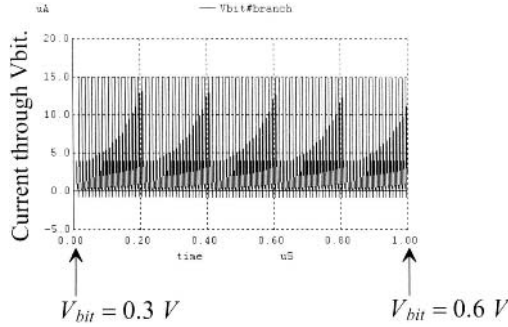


Figure 17.12 How the charge supplied to V_{bit} becomes linear if we add M4, from Fig. 17.9, to the simulation in Fig. 17.10. Note the reduced bit line voltage swing and the reduction in the current pulse amplitudes.

Incomplete Settling

If we zoom in on the current pulses in Fig. 17.12, we get a view like the one seen in Fig. 17.13. The glitches at 30 and 40 ns are the result of the ϕ_2 clock going high and M2 shutting off. Notice how the current through M4 (and V_{bi}) isn't zero when M2 shuts off. Using our water analogy, this is equivalent to stopping the pouring of the water out of the cup before the cup is empty. This incomplete emptying of the cup or capacitor is termed *incomplete settling*. The currents in the circuit haven't gone to zero before the clock transitions. The parasitic capacitances on the sources of M3/M4 continue to discharge through V_{bit} after M2 shuts off, resulting in nonzero current. Incomplete settling has the effect of making the capacitor, C_{cup} , appear smaller. It won't affect the linearity of the sense but it can affect the maximum value of the sensed current. To eliminate the incomplete settling behavior we can increase the width of M4. This results in the source of M4 being held closer to $V_{REF} + V_{THP}$ when M2 and M3 turns on. We could also slow the clock frequency down to ensure that the circuit settles. Because the amount of charge transferred from C_{cup} is constant from one clock cycle to the next, the effects of incomplete settling on DSM are simply a limit on the maximum current that can be removed from the bit line and a modification to the digital output for the absolute value of the input signal, that is, Eq. (17.2) (a smaller value of fed-back signal size is used).

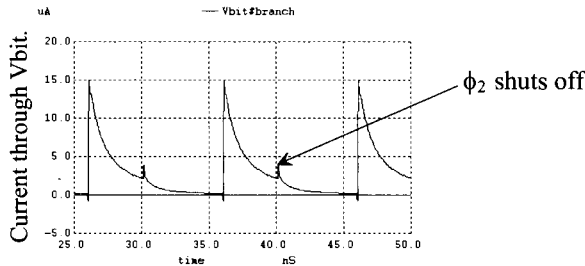


Figure 17.13 The incomplete settling in Fig. 17.12.

17.2 Sensing Resistive Memory

Let's apply the techniques we've just discussed to sensing a resistive memory cell. Figure 17.14a shows a schematic of one-transistor, one-resistor (1T1R) memory cell. Notice the similarity to the 1T1C DRAM memory cell seen in Fig. 16.9. *Ideally* the resistor is either zero ohms (which we'll call *programmed*) or infinite (which we'll call *erased*), Fig. 17.14b. To sense in this ideal case, we simply precharge the bit line to V_{DD} . When the word line is driven high, the access device turns on, resulting in the bit line either moving to $V_{DD}/2$ (if the cell is programmed) or not moving at all (if the cell is erased). In a practical circuit, however, the cell's resistance will be nonzero and not infinite (say $10\text{k}\Omega$ to $100\text{k}\Omega$ as seen in Fig. 17.14c).

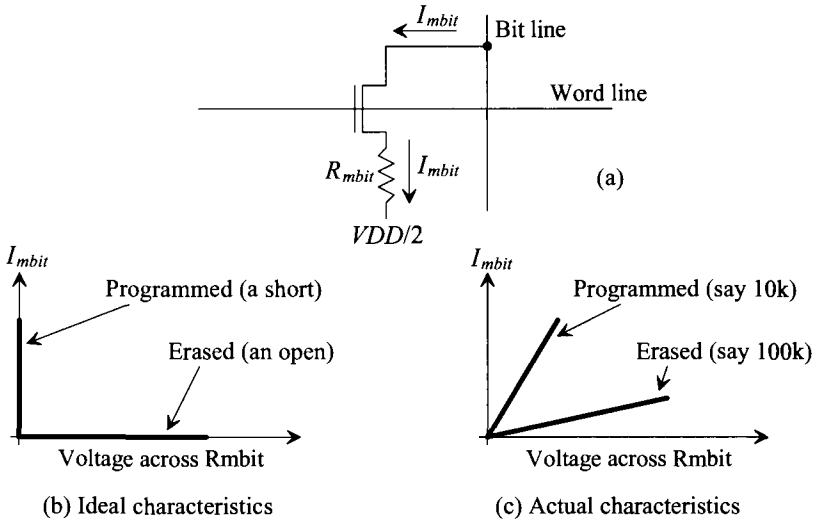


Figure 17.14 The one-transistor, one-resistor (1T1R) RAM memory cell.

The Bit Line Voltage

Consider the block diagram of a DSM seen in Fig. 17.15 with a resistive memory cell. To keep the schematic simpler, we won't include the access MOSFET. We know from the previous discussions that the DSM tries to hold the bit line at precisely the reference voltage used by the comparator. In Fig. 17.15 if the reference voltage used by the comparator is $V_{DD}/2$, then the current that flows through the memory resistor is zero. To avoid this, we might use a reference that is offset from $V_{DD}/2$ by V_{os} . The current that flows in the resistor is then

$$I_{mbit} = \frac{V_{os}}{R_{mbit}} \quad (17.16)$$

In the ideal condition, the erased resistor results in zero I_{mbit} . The output of the DSM is then simply a string of zeroes (we never have to add current to the bit line because none is leaving). If the resistor is programmed, the bitline gets pulled down towards $V_{DD}/2$. The current source can't supply enough charge to pull the bit line up and the DSM output is a constant string of ones (yes, add current).

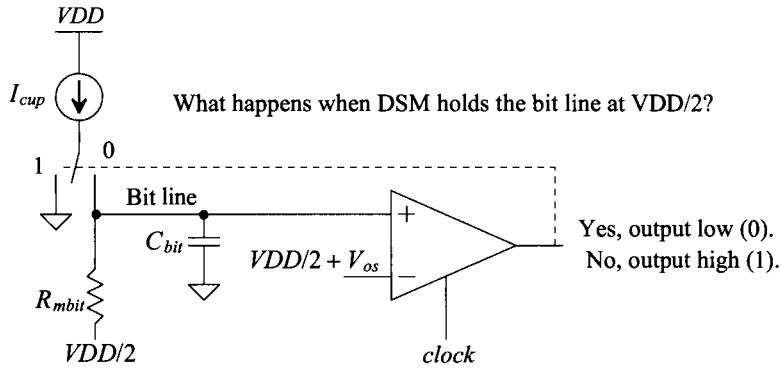


Figure 17.15 Sensing a resistive memory cell using DSM. Notice that we've eliminated the inverter from Fig. 17.8 by switching the input terminals of the comparator.

Adding an Offset to the Comparator

It's important to minimize the number of reference voltages used in a memory sense scheme. We'll use VDD and $VDD/2$ and then design a comparator with a built-in offset voltage. The comparator design is seen in Fig. 17.16 (see Figs. 16.32 and 16.35 from the last chapter). Simulation results showing the offset are seen in Fig. 17.17. When $In+$ gets 50 mV above $In-$, the output of the comparator changes states. It's possible to design the comparator to simplify the sense circuitry, e.g., clock the comparator on the falling edge of ϕ_2 , remove the NAND gates on the output, and combine M2 and M3 in Fig. 17.9. In a

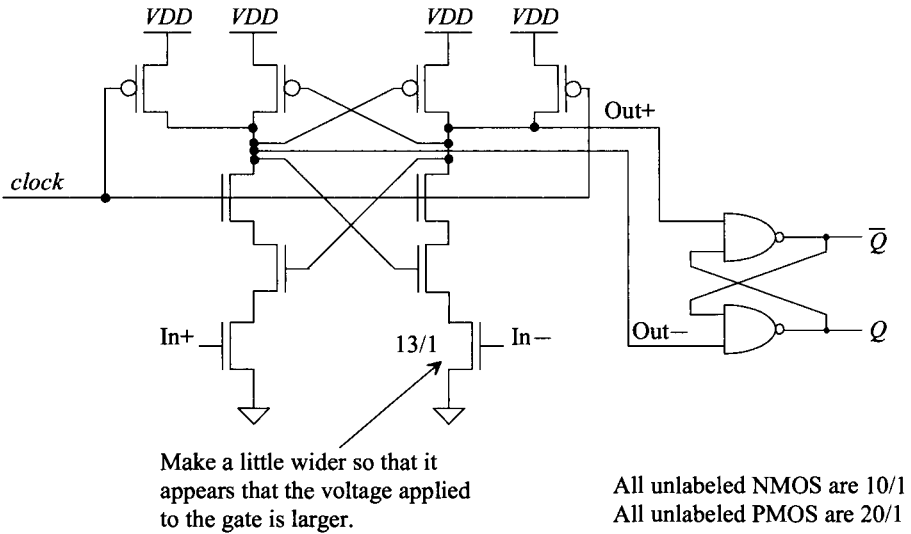


Figure 17.16 Designing a clocked comparator with a built-in offset.